

Roll No.

TBT-304

**B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

BIOCHEMISTRY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss *three* regulatory or key reactions of glycolysis. (CO1)
- (b) Define fermentation. Explain the conversion of pyruvate into ethanol. (CO1)
- (c) Explain the induced fit theory of enzymatic reaction. (CO1)
2. (a) Write down the following reaction steps : (CO1 & CO2)
 - (i) Conversion of Pyruvate to Phosphoenolpyruvate in gluconeogenesis
 - (ii) Conversion of pyruvate to acetyl Co-A

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- (b) (i) How many molecules of ATP or GTP or NADH are produced per molecule of glucose in glycolysis ? (CO1 & CO2)
- (ii) How many molecules of ATP or GTP or NADH are required to the formation of one molecule of glucose from pyruvate ?
- (c) Describe the transamination reaction using an appropriate example. (CO1 & CO2)
3. (a) Draw flow chart of Kreb's cycle with all labelled steps. (CO3)
- (b) Discuss the chemiosmotic theory of ATP synthesis with neat diagram. (CO3)
- (c) (i) Explain the factors on which melting point of a fatty acid depends. (CO3)
- (ii) What are triacylglycerols ? Draw neat diagram of cholesterol. (CO3)
4. (a) Draw any *four* of the following structures : (CO4)
- (i) D-Glucose
- (ii) Phenylalanine
- (iii) ATP
- (iv) Uracil
- (v) Arachidonic acid
- (b) Draw self-explanatory diagram for urea cycle. (CO4)
- (c) Explain the classification enzymes with suitable examples. (CO4)

(3)

5. (a) Describe the glyoxylate pathway of formation of glucose molecule from acetate. (CO4/CO5)
- (b) Explain how Type A, B and Z-DNA differ from one another. (CO4/CO5)
- (c) (i) How many molecules of ATP or GTP or NADH are produced in one cycle of TCA ? (CO4/CO5)
- (ii) How many ATP are required to product one molecule of Urea ?

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**B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

MICROBIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the significance of chemical composition of bacteria cell wall for their identification by Gram's staining. (CO1)
- (b) Give any *one* method of animal virus cultivation using suitable diagram. (CO1)
- (c) Explain any gene transfer mechanism found in bacteria. (CO1)
2. (a) How the biogenesis theory was established ? Give the role of major contributors. (CO2)
- (b) What do you mean by Germ Theory of Disease ? Discuss Koch Postulate. (CO2)
- (c) Explain any method of bacteria classification. (CO2)

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(2)

3. (a) A mixed culture is provided to you. Explain any pure culture technique to isolate pure culture. (CO3)
(b) What are the basic nutritional requirements of bacteria ? How the media are classified on the basis of function ? (CO3)
(c) A soil sample (1g) is provided to you. Explain the method for enumerating bacterial cells as CFU per ml in the given sample. (CO3)
4. (a) Explain the bacterial growth phase using suitable diagram. How the log phase of cells can be obtained ? (CO4)
(b) Using suitable diagram, explain the structure of a bacteriophage. (CO4)
(c) List the factors that influence microbial growth. Explain any *one*. (CO4)
5. (a) How the aseptic conditions are maintained in Laminar Air Flow ? (CO5)
(b) Explain any physical mode of sterilization using moist heating. (CO5)
(c) A contaminated water sample is provided to you. Explain any method of bacterial identification and isolation. (CO5)

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TBT-302

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

CELL BIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Draw a schematic diagram explaining various components of microscope. (CO1)
- (b) Explain the process packaging of DNA into chromosomes. (CO1)
- (c) Write short notes on the following : (CO1)
 - (i) Ribosomes
 - (ii) Polypeptide bond
2. (a) Explain the lipid bi-layer structure with the help of a suitable diagram. (CO2)

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- (b) Write short notes on the following : (CO2)
- (i) Golgi body
 - (ii) Endoplasmic reticulum
- (c) Explain the process of vesicular transport inside a eukaryotic cell. (CO2)
3. (a) Discuss the structure and function of microtubules in detail. (CO3)
- (b) Write short notes on the following : (CO3)
- (i) RTK Signaling
 - (ii) NO Signaling
- (c) What are cell adhesion molecules (CAM) ? Explain the structure and function of CAM. (CO3)
4. (a) Why meiosis is known as reductional division ? Explain with the help of suitable diagram. (CO4)
- (b) Explain the role of cyclins and Cdks in cell cycle regulation. (CO4)
- (c) Give a detailed account of apoptosis in eukaryotic cell. (CO4)
5. (a) Highlight the importance of any *two* model organisms in detail. (CO5)
- (b) Write short notes on the following : (CO5)
- (i) FISH
 - (ii) BLAST
- (c) Name any *five* staining dyes used in cell biology and briefly explain their functions. (CO5)

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TBT-301

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

GENETICS AND MOLECULAR BIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Compare Quantitative and Qualitative Genetics in *ten* points.
(b) Write a short note on Epistasis. Explain epistatic and hypostatic gene with examples.
(c) Briefly explain the Lethal genes with an example from plants, animals and humans.
2. (a) Draw the structure of all the nitrogenous bases found in DNA and RNA.
(b) Illustrate DNA Packaging in Eukaryotic cells.
(c) Exemplify the behaviour of DNA with radiation and temperature.

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(2)

3. (a) Briefly explain how the cells survive from various kinds of DNA damages.
(b) Write a short note on DNA Replication Enzymes and their function.
(c) Illustrate the transcription process in Eukaryotes.
4. (a) What is the genetic code ? Describe the characteristic features of genetic code.
(b) Pen down the post-translational modifications.
(c) Illustrate the process of End modifications of RNA in eukaryotic RNA processing.
5. (a) Discuss how the trp operon is controlled in Prokaryotes.
(b) How gene expression in Eukaryotic cells is regulated.
(c) Define Gene silencing. What are their applications in heartcare ?